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```
import numpy as np
#for and gate (input/output)
x_list = np.array([[0,0],[0,1],[1,0],[1,1]])
y_list = np.array([0,0,0,1])
epochs = 4
w,b = [1,2], -1
for _ in range(epochs):
    for x,y in zip(x_list,y_list):
        z = np.dot(x,w) + b
        y_pred = 1 if z >=0 else 0
        E = y-y_pred
        print("error:",E,"<-at epoch",_)
        w += E*x
        b += E
    print(w,b)
```

```
error: 0 <-at epoch 0
[1 2] -1
error: -1 <-at epoch 0
[1 1] -2
error: 0 <-at epoch 0
[1 1] -2
error: 0 <-at epoch 0
[1 1] -2
error: 0 <-at epoch 1
[1 1] -2
error: 0 <-at epoch 1
[1 1] -2
error: 0 <-at epoch 1
[1 1] -2
error: 0 <-at epoch 1
[1 1] -2
error: 0 <-at epoch 2
[1 1] -2
error: 0 <-at epoch 2
[1 1] -2
error: 0 <-at epoch 2
[1 1] -2
error: 0 <-at epoch 2
[1 1] -2
error: 0 <-at epoch 3
[1 1] -2
error: 0 <-at epoch 3
[1 1] -2
error: 0 <-at epoch 3
[1 1] -2
error: 0 <-at epoch 3
[1 1] -2
```

```
import numpy as np
#for and gate (input/output)
x_list = np.array([[0,0],[0,1],[1,0],[1,1]])
y_list = np.array([0,0,0,1])
epochs = 6
w,b = np.zeros(2),0
for _ in range(epochs):
    for x,y in zip(x_list,y_list):
        z = np.dot(x,w) + b
        y_pred = 1 if z >=0 else 0
        E = y-y_pred
        print("error:",E,"<-at epoch",_)
        w += E*x
        b += E
    print(w,b)
```

```
error: -1 <-at epoch 0
[0. 0.] -1
error: 0 <-at epoch 0
[0. 0.] -1
error: 0 <-at epoch 0
[0. 0.] -1
error: 1 <-at epoch 0
[1. 1.] 0
error: -1 <-at epoch 1
[1. 1.] -1
error: -1 <-at epoch 1
[1. 0.] -2
error: 0 <-at epoch 1
[1. 0.] -2
error: 1 <-at epoch 1
[2. 1.] -1
error: 0 <-at epoch 2
[2. 1.] -1
error: -1 <-at epoch 2
[2. 0.] -2
error: -1 <-at epoch 2
[1. 0.] -3
error: 1 <-at epoch 2
[2. 1.] -2
error: 0 <-at epoch 3
[2. 1.] -2
error: 0 <-at epoch 3
[2. 1.] -2
error: -1 <-at epoch 3
[1. 1.] -3
error: 1 <-at epoch 3
[2. 2.] -2
error: 0 <-at epoch 4
[2. 2.] -2
error: -1 <-at epoch 4
```

```
[2. 1.] -3
error: 0 <-at epoch 4
[2. 1.] -3
error: 0 <-at epoch 4
[2. 1.] -3
error: 0 <-at epoch 5
[2. 1.] -3
error: 0 <-at epoch 5
[2. 1.] -3
error: 0 <-at epoch 5
[2. 1.] -3
error: 0 <-at epoch 5
[2. 1.] -3
```

```
import numpy as np
#for and gate (input/output)
x_list = np.array([[0,0],[0,1],[1,0],[1,1]])
y_list = np.array([0,0,0,1])
epochs = 5
w,b = np.zeros(2),0
for _ in range(epochs):
    for x,y in zip(x_list,y_list):
        z = np.dot(x,w) + b
        y_pred = 1 if z >=0 else 0
        E = y-y_pred
        print("error:",E,"<-at epoch",_)
        w += E*x
        b += E
    print(w,b)
```

```
error: -1 <-at epoch 0
[0. 0.] -1
error: 0 <-at epoch 0
[0. 0.] -1
error: 0 <-at epoch 0
[0. 0.] -1
error: 1 <-at epoch 0
[1. 1.] 0
error: -1 <-at epoch 1
[1. 1.] -1
error: -1 <-at epoch 1
[1. 0.] -2
error: 0 <-at epoch 1
[1. 0.] -2
error: 1 <-at epoch 1
[2. 1.] -1
error: 0 <-at epoch 2
[2. 1.] -1
error: -1 <-at epoch 2
[2. 0.] -2
error: -1 <-at epoch 2
[1. 0.] -3
error: 1 <-at epoch 2
[2. 1.] -2
error: 0 <-at epoch 3
[2. 1.] -2
error: 0 <-at epoch 3
[2. 1.] -2
error: -1 <-at epoch 3
[1. 1.] -3
error: 1 <-at epoch 3
[2. 2.] -2
error: 0 <-at epoch 4
[2. 2.] -2
error: -1 <-at epoch 4
[2. 1.] -3
error: 0 <-at epoch 4
[2. 1.] -3
error: 0 <-at epoch 4
[2. 1.] -3
```

```
import numpy as np
# for AND gate (Input/Output)
x_list = np.array([[0,0],[0,1],[1,0],[1,1]])
y_list = np.array([0,0,0,1])
epochs = 10
w,b = [2,1],0

for _ in range(epochs):
    total_error_in_epoch = 0
    for x,y in zip(x_list,y_list):
        z = np.dot(x,w) + b
        # condition area for y_pred
        y_pred = 1 if z >= 0 else 0
        E = y - y_pred
        total_error_in_epoch += abs(E)
        print("error : ",E,"<- at epoch",_)
        w += E*x
        b += E
    print(w,b)
    if total_error_in_epoch == 0:
        print(f"Converged at epoch {_}. No errors found.")
        break
```

```
error : -1 <- at epoch 0
[2 1] -1
error : -1 <- at epoch 0
[2 0] -2
error : -1 <- at epoch 0
[1 0] -3
error : 1 <- at epoch 0
[2 1] -2
error : 0 <- at epoch 1
[2 1] -2
error : 0 <- at epoch 1
[2 1] -2
error : -1 <- at epoch 1
```

```

[1 1] -3
error : 1 <- at epoch 1
[2 2] -2
error : 0 <- at epoch 2
[2 2] -2
error : -1 <- at epoch 2
[2 1] -3
error : 0 <- at epoch 2
[2 1] -3
error : 0 <- at epoch 2
[2 1] -3
error : 0 <- at epoch 3
[2 1] -3
error : 0 <- at epoch 3
[2 1] -3
error : 0 <- at epoch 3
[2 1] -3
error : 0 <- at epoch 3
[2 1] -3
Converged at epoch 3. No errors found.

```

```

import numpy as np
# for OR gate (Input/Output)
x_list = np.array([[0,0],[0,1],[1,0],[1,1]])
y_list = np.array([0,1,1,1]) # Changed y_list for OR gate
epochs = 10
w,b = [1,2],1

for _ in range(epochs):
    total_error_in_epoch = 0
    for x,y in zip(x_list,y_list):
        z = np.dot(x,w) + b
        ### condition area for y_pred
        y_pred = 1 if z >= 0 else 0
        E = y - y_pred
        total_error_in_epoch += abs(E)
        print("error : ",E,"<- at epoch",_)
        w += E*x
        b += E
    print(w,b)
    if total_error_in_epoch == 0:
        print(f"Converged at epoch {_}. No errors found.")
        break

```

```

error : -1 <- at epoch 0
[1 2] 0
error : 0 <- at epoch 0
[1 2] 0
error : 0 <- at epoch 0
[1 2] 0
error : 0 <- at epoch 0
[1 2] 0
error : -1 <- at epoch 1
[1 2] -1
error : 0 <- at epoch 1
[1 2] -1
error : 0 <- at epoch 1
[1 2] -1
error : 0 <- at epoch 1
[1 2] -1
error : 0 <- at epoch 2
[1 2] -1
error : 0 <- at epoch 2
[1 2] -1
error : 0 <- at epoch 2
[1 2] -1
error : 0 <- at epoch 2
[1 2] -1
Converged at epoch 2. No errors found.

```